BADJI MOKHTAR UNIVERSITY ANNABA

FACULTY OF MEDICINE

FIRST YEAR OF MEDICINE

PHYSIOLOGY

THE AUTONOMIC NERVOUS SYSTEM

Student Course Handout

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Learning objectives

By the end of this lecture, you should be able to:

1. Identify the targets and functions of the autonomic nervous system.
2. Compare the origins, pathways and effects of the sympathetic and parasympathetic divisions.
3. Describe the usual two-neuron autonomic motor pathway.
4. Match the major neurotransmitters with their receptors.
5. Explain short (local) and long (central) visceral reflexes.

> **Colour code used in all diagrams**
>
> **Orange** = sympathetic division · **Green** = parasympathetic division · Purple boxes = central nervous system (CNS)
>
> **ACh** = acetylcholine (pink boxes) · **NE** = noradrenaline / norepinephrine (blue boxes)
>
> Solid line —— = preganglionic fibre · Dashed line ┈┈ = postganglionic fibre

1. Overview and homeostasis

The **autonomic nervous system (ANS)** is a functional division of the nervous system that regulates **visceral effectors**: cardiac muscle, smooth muscle and glands. Blood vessels are major targets because their walls contain smooth muscle. It is traditionally classified as part of the peripheral nervous system, although its control centres lie in the CNS.

Autonomic activity is **predominantly involuntary**, but it is not independent of the brain: emotions, exercise, sleep and breathing all change it. "Autonomic" means automatic, not autonomous from the brain.

Somatic versus autonomic motor systems

Feature	Somatic motor system	Autonomic motor system
Targets	Skeletal muscle	Cardiac muscle, smooth muscle, glands
Efferent pathway	One neuron from CNS to muscle	Usually two neurons, relayed in a peripheral ganglion
Transmitter at target	**ACh** on nicotinic muscle receptors (Nm)	**ACh** or **NE**, depending on the pathway
Effect on target	Always excitatory	Excitatory or inhibitory
Control	Voluntary movement and reflexes	Predominantly involuntary

Visceral sensory (afferent) neurons report internal conditions (stretch, pressure, chemistry). **Autonomic motor (efferent)** pathways carry the commands to the effectors.

Diagram 1. The four main homeostatic jobs of the ANS

Cardiovascular Adjusts heart rate and vessel tone, supporting blood pressure and tissue perfusion.	**Digestive** Adjusts gut motility and secretion, supporting nutrient processing.
Thermoregulation Sweating and skin blood flow help regulate body temperature.	**Bladder** Coordinates urine storage and emptying.

Homeostasis is continuous adjustment, not a fixed internal state.

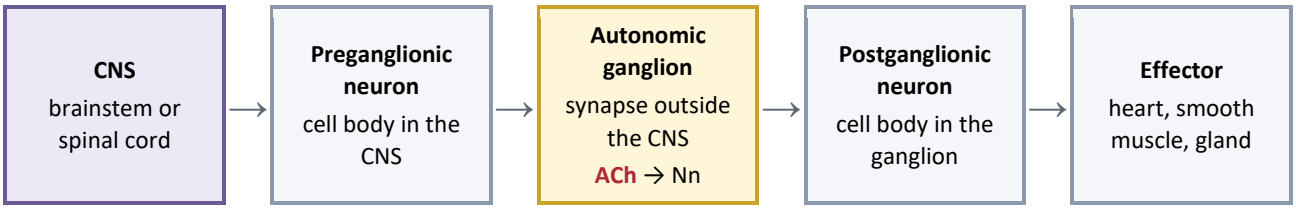
2. Organization of autonomic pathways

The ANS has three components:

- **Sympathetic division**: thoracolumbar origin; mobilizes the body for activity and stress ("fight or flight").
- **Parasympathetic division**: craniosacral origin; supports digestion, secretion, elimination and recovery ("rest and digest").
- **Enteric nervous system**: an intrinsic network in the gut wall. It can organize local activity on its own and is modulated by both divisions.

The usual two-neuron pathway

Diagram 2. From the CNS to the effector: two neurons, one ganglion

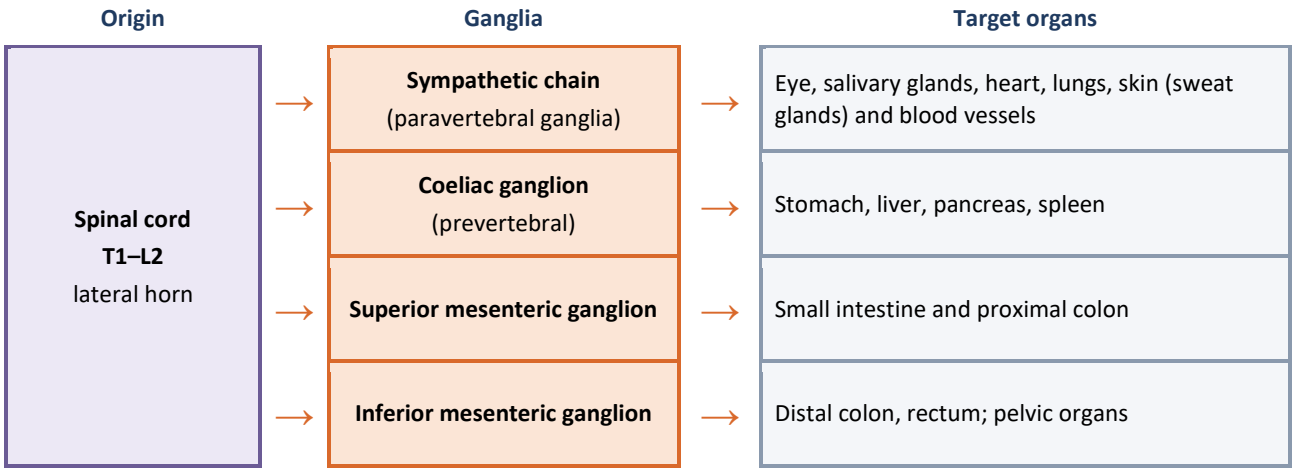


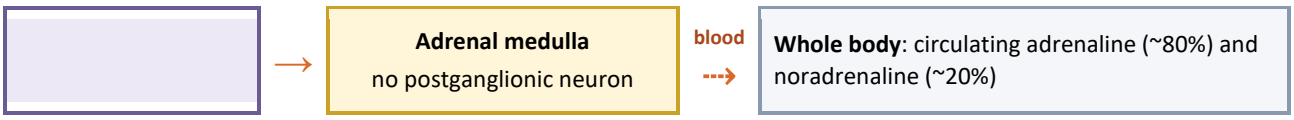
A ganglion is a cluster of neuronal cell bodies outside the CNS. The adrenal medulla is the main exception to this pattern (see below).

Sympathetic outflow (thoracolumbar)

Sympathetic preganglionic neurons lie in the lateral horn of the spinal cord from **T1 to L2**. Their fibres follow one of three routes:

Diagram 3. Sympathetic pathways: origin → ganglia → target organs





Routes 1–2: fibres synapse in the chain or in prevertebral ganglia near the large abdominal arteries. Route 3: the preganglionic fibre reaches the adrenal medulla directly.

The **adrenal medulla** behaves like a modified sympathetic ganglion. Its **chromaffin cells** (modified postganglionic neurons) are stimulated by preganglionic **ACh** through Nn receptors and release hormones into the blood instead of sending an axon to an organ.

Parasympathetic outflow (craniosacral)

Parasympathetic preganglionic neurons lie in the **brainstem** (cranial nerves III, VII, IX, X) and in the **sacral spinal cord (S2–S4)**. Their ganglia lie close to or inside the target organ, which allows very local control.

Diagram 4. Parasympathetic pathways: origin → nerve → ganglion → target

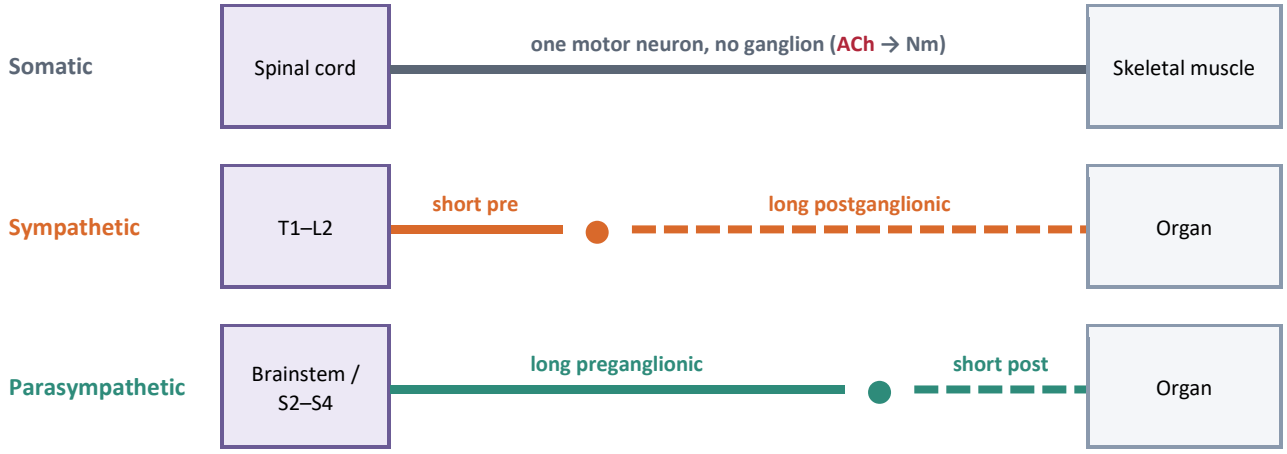
Origin	Nerve	Ganglion	Target organs
Brainstem (cranial part)	CN III →	**Ciliary** ganglion	Eye: pupil constriction and accommodation (lens)
	CN VII →	**Pterygopalatine** ganglion	Lacrimal and nasal glands
	CN VII →	**Submandibular** ganglion	Submandibular and sublingual salivary glands
	CN IX →	**Otic** ganglion	Parotid salivary gland
	CN X (vagus) →	Ganglia near or within organs	Heart, lungs, oesophagus, stomach, small intestine, proximal colon
Sacral cord S2–S4	Pelvic splanchnic nn. →	Ganglia near or within organs	Distal colon, rectum, bladder, erectile tissue

The vagus nerve (X) carries most parasympathetic outflow: it supplies the thoracic organs and most of the digestive tract.

Comparing the two divisions

Feature	Sympathetic	Parasympathetic
CNS origin	T1–L2 (thoracolumbar)	Brainstem and S2–S4 (craniosacral)
Ganglia	Near the vertebral column (chain) or major abdominal arteries (prevertebral)	Near or within the target organs
Preganglionic fibre	Usually **short**	Usually **long**
Postganglionic fibre	Usually **long**	Usually **short**
Divergence	High: one preganglionic neuron reaches many ganglia, giving widespread effects	Low: localized, organ-specific effects

Diagram 5. Where the ganglion sits decides the fibre lengths



● = *autonomic ganglion. These are general patterns; there are anatomical exceptions.*

Key points

Sympathetic = **thoracolumbar**, ganglia **far** from the organ (short pre, long post).

Parasympathetic = **craniosacral**, ganglia **near or in** the organ (long pre, short post).

The adrenal medulla receives a preganglionic fibre directly and releases hormones into the blood.

3. Neurotransmitters and receptors

Two rules cover almost every case:

1. **In every autonomic ganglion**, the preganglionic neuron releases **ACh**, which acts on **neuronal nicotinic (Nn)** receptors. This is true for both divisions.
2. **At the target organ**, parasympathetic fibres release **ACh** (muscarinic receptors) and most sympathetic fibres release **NE** (adrenergic α or β receptors).

Diagram 6. Transmitter and receptor at each step of the pathway

	Parasympathetic	Sympathetic (most organs)	Sympathetic → sweat glands	Adrenal medulla
Preganglionic neuron releases	ACh	ACh	ACh	ACh
↓	↓	↓	↓	↓
Receptor in the ganglion	Nn	Nn	Nn	Nn on chromaffin cells
↓	↓	↓	↓	↓
Postganglionic neuron releases	ACh	NE	ACh (exception)	No neuron: adrenaline + NE into blood
↓	↓	↓	↓	↓
Receptor on target cell	Muscarinic (M)	Adrenergic α or β	Muscarinic (M_3)	α or β (via blood)
↓	↓	↓	↓	↓
Example	Heart slows (M_2)	Vessels constrict (α_1); heart speeds up (β_1)	Sweating	Whole-body stress response

Read each column from top to bottom. Pink = cholinergic step (ACh), blue = adrenergic step (NE / adrenaline).

Exceptions and special cases to remember

1. Eccrine sweat glands: sympathetic, but **cholinergic**. Postganglionic fibres release ACh onto M_3 receptors. This is the key exception.

2. Adrenal medulla: no postganglionic neuron. Preganglionic ACh → Nn on chromaffin cells → adrenaline (~80%) and noradrenaline (~20%) released into the blood.

3. Kidney (β_1 → renin) and adipose tissue (β_3 → lipolysis) are *not* exceptions: they use NE normally, through a particular receptor subtype.

4. Bronchi and some vascular beds depend mainly on **circulating adrenaline** acting on β_2 receptors, because they receive little direct sympathetic innervation.

Note: sympathetic cholinergic vasodilator fibres to skeletal muscle are well documented in some animals, but their role in humans is debated.

Receptor signalling

Receptor	Main mechanism	Main locations (examples)
Nn	Ligand-gated cation channel: fast, no second messenger	All autonomic ganglia; adrenal medulla
M_1, M_3, M_5	Gq → IP_3/DAG → ↑ intracellular Ca^{2+}	M_3: smooth muscle (gut, bronchi, bladder, iris), glands, endothelium
M_2, M_4	Gi/o → ↓ cAMP; M_2 also opens K^+ channels	M_2: heart (SA and AV nodes, atria)
α_1	Gq → IP_3/DAG → ↑ Ca^{2+}	Vascular smooth muscle, iris dilator, sphincters
α_2	Gi/o → ↓ cAMP	Presynaptic terminals (reduce NE release), pancreatic β-cells
β_1, β_2, β_3	Mainly Gs → ↑ cAMP	β_1: heart, kidney · β_2: bronchi, some vessels, liver · β_3: adipose tissue, detrusor

Memory aid for G proteins

Gs = **S**timulates (↑ cAMP) · **Gi** = **I**nhibits (↓ cAMP) · **Gq** = Ca^{2+} (IP_3/DAG)

The effect also depends on the tissue: M_3 **contracts** the iris sphincter, but endothelial M_3 **relaxes** vascular smooth muscle by releasing nitric oxide (NO).

4. Visceral reflexes and central control

A **visceral reflex** starts when a receptor detects a change. Sensory pathways carry the information to an integrating circuit, which produces a motor response in an effector. There are two types:

- **Short reflex**: integrated in peripheral circuits (enteric plexus or autonomic ganglion) **without** passing through the CNS. Fast and local: it affects a small part of one organ.
- **Long reflex**: visceral afferents reach the spinal cord or brainstem, and autonomic efferents carry the response back. It can coordinate several organs.

Diagram 7. Short versus long visceral reflex

Short reflex (local)	Long reflex (central)
Stimulus: stretch of the gut wall	**Stimulus**: bladder filling
↓	↓
Receptor in peripheral tissue	Receptor in peripheral tissue
↓	↓
Afferent (sensory) fibre	Visceral sensory neuron
↓	↓
No CNS involvement	**CNS integrating centre**: spinal cord or brainstem (interneurons)
↓	↓
Integration in the **autonomic ganglion** or **enteric plexus**	Preganglionic neuron → autonomic ganglion
↓	↓
Ganglionic (postganglionic) neuron	Postganglionic neuron
↓	↓
Effector → local response (e.g. nearby contraction)	**Effector** → coordinated response (e.g. urination)

Main difference: the short reflex stays local, the long reflex passes through the CNS (dashed box = step absent).

Central regulation

The ANS is controlled by a **hierarchy** of centres. Higher levels adapt autonomic responses to emotional and physiological context; this is why exam stress can speed the heart and slow digestion.

Diagram 8. Central control of the ANS, from top to bottom

Cortex and limbic system
Emotions, thoughts and behaviour

↓

Hypothalamus
Main coordinator of autonomic responses and homeostasis (temperature, hunger, thirst)

↓

Brainstem
Vital reflexes: heart rate, blood pressure, breathing

↓

Spinal cord
Reflexes for bladder and bowel emptying

↓

Autonomic nerves → internal organs

Vagus nerve (X): direct route from the brainstem to the heart, lungs and digestive tract

↓

5. Effects on major organs

Notation: effect, then **transmitter → receptor**. NE = noradrenaline; Adr = circulating adrenaline; ACh = acetylcholine.

Organ	Sympathetic	Parasympathetic
Heart	↑ rate, ↑ AV conduction, ↑ contractility: NE → β_1	↓ rate, ↓ AV conduction: ACh → M_2
Blood vessels	Most vessels: vasoconstriction, NE → α_1. Skeletal-muscle vessels: vasodilation, Adr → β_2	Most vessels: no direct innervation
Bronchi	Bronchodilation: mainly Adr → β_2	Bronchoconstriction, ↑ secretion: ACh → M_3
Eye: iris	Pupil dilation (radial dilator muscle): NE → α_1	Pupil constriction (sphincter): ACh → M_3
Eye: ciliary muscle	Slight relaxation (far vision): β_2	Contraction, accommodation for near vision: ACh → M_3
Salivary glands	Scant, protein-rich secretion: α_1, β	Abundant watery secretion: ACh → M_3
Lacrimal glands	Little effect	↑ tear secretion: ACh → M_3
Gut	↓ motility and secretion; sphincters contract: NE → α, β	↑ motility and secretion; sphincters relax: ACh → M_3
Liver	↑ glycogenolysis and gluconeogenesis: Adr/NE → β_2, α_1	Little direct effect
Pancreas	↓ insulin secretion: NE → α_2	↑ insulin and enzyme secretion: ACh → M_3
Adipose tissue	↑ lipolysis: NE → β_3	No innervation
Kidney	↑ renin release: NE → β_1	No innervation
Adrenal medulla	Secretes adrenaline + NE into blood: ACh → Nn	No innervation
Eccrine sweat glands	↑ sweating: ACh → M_3 (cholinergic exception)	No innervation
Bladder: detrusor	Relaxation (storage): NE → β_3	Contraction (emptying): ACh → M_3
Bladder: internal sphincter	Contraction (storage): NE → α_1	Relaxation (emptying)
Uterus	Relaxation (β_2) or contraction (α_1), depending on hormonal state	Variable, minor
Male sexual function	Emission and ejaculation: NE → α_1	Erection (vasodilation): ACh → M, via NO
Female sexual function	Rhythmic smooth-muscle contraction at orgasm: NE → α_1	Vasodilation, clitoral erection and lubrication: via NO

Memory aid for sexual function: "Point and Shoot"

Parasympathetic = **P**oint (erection) · **S**ympathetic = **S**hoot (emission and ejaculation).

Opposition and cooperation

Type of interaction	Examples
Opposing effects	Heart rate: sympathetic ↑, parasympathetic ↓. Pupil: dilation versus constriction. Bladder: storage versus emptying.
Complementary effects	Salivation: both divisions stimulate secretion, with different composition. Sexual function: erection (parasympathetic) then ejaculation (sympathetic).
Single innervation	Most blood vessels, sweat glands, adrenal medulla and kidney receive only sympathetic fibres: responses come from ↑ or ↓ sympathetic tone.

6. Key takeaways

- Autonomic efferents regulate visceral effectors: cardiac muscle, smooth muscle and glands.
- Sympathetic outflow is **thoracolumbar** (T1–L2); parasympathetic outflow is **craniosacral** (CN III, VII, IX, X and S2–S4).
- Most pathways contain a **preganglionic** and a **postganglionic** neuron, relayed in a ganglion.
- Both divisions use **ACh on Nn receptors** in the ganglia.
- At the target: parasympathetic **ACh → M**; sympathetic usually **NE → α/β**; sweat glands are the cholinergic exception.
- The organ response depends on transmitter, receptor subtype and tissue.
- Short (local) and long (central) reflexes, under hypothalamic and cortical influence, coordinate visceral regulation.

Abbreviations

Abbr.	Meaning	Abbr.	Meaning
ACh	Acetylcholine	**CN**	Cranial nerve
NE	Noradrenaline (norepinephrine)	**SA, AV**	Sinoatrial node, atrioventricular node
Adr	Adrenaline (epinephrine)	**cAMP**	Cyclic adenosine monophosphate
Nn / Nm	Nicotinic receptor: neuronal / muscle type	**IP_3, DAG**	Inositol trisphosphate, diacylglycerol
M_1–M_5	Muscarinic receptor subtypes	**NO**	Nitric oxide

The Autonomic Nervous System

English–Arabic terminology | First-year medical physiology

Arabic equivalents are provided for the meaning used in this course.

Key terms

English	العربية
Autonomic nervous system	الجهاز العصبي الذاتي (اللاإرادي)
Central nervous system	الجهاز العصبي المركزي
Peripheral nervous system	الجهاز العصبي المحيطي
Sympathetic nervous system	الجهاز العصبي الودّي
Parasympathetic nervous system	الجهاز العصبي نظير الودّي
Enteric nervous system	الجهاز العصبي المعوي
Visceral organ	عضو حشوي
Effector	عضو مستجيب
Cardiac muscle	العضلة القلبية
Smooth muscle	العضلات الملساء
Gland	غدة
Homeostasis	الاستتباب
Afferent pathway	مسار وارد حسّي
Efferent pathway	مسار صادر حركي
Preganglionic neuron	عصبون قبل عقدي
Postganglionic neuron	عصبون بعد عقدي
Autonomic ganglion	عقدة عصبية ذاتية
Sympathetic chain	السلسلة الودّية
Prevertebral ganglion	عقدة أمام الفقرات
Thoracolumbar outflow	المنشأ الصدري القطني
Craniosacral outflow	المنشأ القحفي العجزي
Brainstem	جذع الدماغ
Spinal cord	النخاع الشوكي
Cranial nerve	عصب قحفي
Vagus nerve	العصب المبهم
Pelvic splanchnic nerves	الأعصاب الحشوية الحوضية
Adrenal medulla	لبّ الغدة الكظرية
Chromaffin cell	خلية كرومافينية
Neurotransmitter	ناقل عصبي
Acetylcholine	أستيل كولين
Norepinephrine / noradrenaline	نورإبينفرين / نورأدرينالين
Epinephrine / adrenaline	إبينفرين / أدرينالين
Nicotinic receptor	مستقبل نيكوتيني
Muscarinic receptor	مستقبل مسكاريني
Adrenergic receptor	مستقبل أدرينرجي
Alpha receptor	مستقبل ألفا
Beta receptor	مستقبل بيتا
Second messenger	رسول ثانٍ
Cyclic AMP (cAMP)	أحادي فوسفات الأدينوزين الحلقي
Intracellular calcium	الكالسيوم داخل الخلية
Visceral reflex	منعكس حشوي

Short reflex	منعكس قصير موضعي
Long reflex	منعكس طويل مركزي
Hypothalamus	الوطاء (تحت المهاد)
Limbic system	الجهاز الحوفي
Heart rate	معدّل ضربات القلب
Contractility	قابلية الانقباض
Blood pressure	ضغط الدم
Tissue perfusion	تروية الأنسجة
Vasoconstriction	تضيّق الأوعية
Vasodilation	توسّع الأوعية
Bronchoconstriction	تضيّق القصبات
Bronchodilation	توسّع القصبات
Pupil dilation / mydriasis	توسّع الحدقة
Pupil constriction / miosis	تضيّق الحدقة
Sweat gland	غدة عرقية
Salivary gland	غدة لعابية
Gastrointestinal motility	حركية الجهاز الهضمي
Bladder	المثانة
Detrusor muscle	العضلة النافصة للمثانة
Sphincter	مَصَرّة
Urination / micturition	التبوّل
Renin	الرينين
Lipolysis	تحلّل الدهون
Erection	الانتصاب
Ejaculation	القذف

Useful verbs

English	العربية
To innervate	يُعصّب
To regulate	ينظّم
To stimulate	ينبّه
To inhibit	يثبّط
To release	يحرّر أو يفرز
To bind to a receptor	يرتبط بمستقبل
To activate	ينشّط
To increase	يزيد
To decrease	يقلّل
To contract	ينقبض
To relax	يرتخي
To constrict	يضيّق
To dilate	يوسّع
To secrete	يفرز
To store	يخزّن
To empty	يفرغ
To coordinate	ينسّق